SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)
Department of Artificial Intelligence and Data Science
ASSESSMENT TOOL 1 – Constraint-Based Problem Solving

Course Code:	DSA03	**Course Title:**	Natural Language Processing
CO Assessed:	CO5 – Articulation	**Assessment:**	Assessment Tool 1 – Constraint-Based Problem Solving
Weightage:	35% of CIA		
CO5 Statement:	Implement semantic models using predicate logic, word sense disambiguation and validate information retrieval techniques. (BL-5)		
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Section / Batch:	D/2024	**Date:**	29.08.26

Code of Conduct

I _AB Dhanyashree(192472002)__ (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: ________________________

Instructions

- Draw necessary diagrams such as constraint graphs, coreference chains, discourse trees, or predicate logic representations wherever applicable.
- Generate solutions that fully satisfy all given constraints and provide clear evaluation of the output.
- Answers should be concise, well-structured, and supported by evidence from the given text/scenario.
- Duration: 30 minutes.

Section / Topic	Focus	Question No.
Discourse, Reference Resolution & Text Coherence	Apply constraint-based problem solving for reference resolution, identify agreement and coherence constraints, resolve coreferences, and ensure entity chain consistency in a given paragraph.	Q1
Dialog and Conversational Agents, Discourse Planning & Surface Realizations	Construct constrained dialog responses by applying dialog acts, discourse planning, coherence and politeness constraints, generate multiple valid responses, and evaluate the best output.	Q2
Machine Translation, Discourse Structure, Predicate Logic & Interpretation	Perform word sense disambiguation using constraints, construct predicate logic representations, maintain discourse structure (rhetorical relations), and generate coherent translations/paraphrases.	Q3

Annexure A – Constraint-Based Problem Solving

1. Reference Resolution and Discourse Coherence

Given the following paragraph:

"Ravi met Arun at the library. He was looking for a book on Artificial Intelligence. Arun helped him find it. Later, they discussed the book before leaving."

Tasks:

a) **Identify all the referring expressions and their possible antecedents.**

b) Apply the following constraints to resolve the references:

- **Gender and number agreement**
- **Recency**
- **Grammatical role**
- **Semantic compatibility**
- **Discourse coherence**

c) Construct a table showing:

Referring Expression Possible Antecedents Valid/Invalid Reason

d) Provide the final coreference chains.

e) Rewrite the paragraph by replacing pronouns with their correct references.

f) Discuss what ambiguity would arise if the sentence "He was looking for a book" appeared before introducing Arun.

2. Dialog Act Recognition and Response Generation

Conversation History:

Student: "I submitted my assignment, but I am worried that I may have made many mistakes."

Required Dialog Acts:

Reassure + Advise

Constraints:

1. **Maintain entity coherence using assignment, mistakes, and you.**
2. **Use a Cause–Effect or Elaboration discourse relation.**
3. **Include at least two keywords from:**
    - **review**
    - **improve**
    - **confident**

- feedback

4. **Response length must be 2–3 sentences.**
5. **Maintain a positive and polite tone.**
6. **Do not give contradictory advice.**

Tasks:

a) Identify the dialog acts required.

b) Identify the important entities that must be maintained.

c) Generate three possible responses satisfying all constraints.

d) Compare the responses based on:

- **Coherence**
- **Constraint satisfaction**
- **Politeness**
- **Relevance**

e) Select the best response and justify your answer.

f) Explain what happens if entity coherence is not maintained.

3. Word Sense Disambiguation and Meaning Representation

Source Sentence:

"Priya sat on the bank and watched the boats moving across the water."

Note:

The word "bank" can refer to:

- **A financial institution**
- **The side of a river**

Constraints:

1. **Resolve the correct word sense using contextual information.**
2. **Identify important entities and actions.**
3. **Represent the meaning using predicate logic.**
4. **Preserve semantic relationships.**
5. **Maintain coherence in the generated paraphrase.**

Tasks:

a) Identify the possible meanings of bank.

b) Select the correct sense and justify your decision using contextual constraints.

c) Write a predicate logic representation using suitable predicates such as:

sit(x)

bank(y)

beside(x,y)

watch(x,z)

boat(z)

move(z)

water(w)

d) Generate a simple English paraphrase without ambiguity.

e) Explain how Word Sense Disambiguation improves Machine Translation.

<u>ANSWER</u>

Q1. Reference Resolution and Discourse Coherence
Given Paragraph:

"Ravi met Arun at the library. He was looking for a book on Artificial Intelligence. Arun helped him find it. Later, they discussed the book before leaving."

a) Referring Expressions and Possible Antecedents

Referring Expression	Possible Antecedents
He	Ravi, Arun
him	Ravi, Arun
it	Book on Artificial Intelligence
they	Ravi and Arun
the book	Book on Artificial Intelligence

b) Applying Constraints

1. **Gender and Number Agreement:** Ravi and Arun are both singular and male, so "He" can refer to either.
2. **Recency:** Arun is mentioned immediately before "He", so recency favors Arun.
3. **Grammatical Role:** Ravi is the subject of the previous sentence, which supports Ravi as the continuing discourse entity.
4. **Semantic Compatibility:** "Looking for a book" is possible for either Ravi or Arun.
5. **Discourse Coherence:** The paragraph is more coherent when Ravi is treated as the main discourse subject and Arun subsequently helps him.

Therefore, the best interpretation is **He → Ravi**.

c) Constraint Table

Referring Expression	Possible Antecedents	Valid/Invalid	Reason
He	Ravi	Valid	Subject continuity and discourse coherence
He	Arun	Less preferred	Recency supports it, but subject continuity is weaker
him	Ravi	Valid	Arun helps Ravi find the book
him	Arun	Invalid/less likely	Arun helping himself is semantically unnatural
it	Book	Valid	"It" refers to the previously mentioned book
they	Ravi + Arun	Valid	Plural pronoun refers to both people
the book	Book on AI	Valid	Definite noun phrase refers to the previously introduced book

d) Final Coreference Chains

- **Ravi → He → him → they**
- **Arun → they**
- **Book on Artificial Intelligence → it → the book**

e) Rewritten Paragraph

“Ravi met Arun at the library. **Ravi** was looking for a book on Artificial Intelligence. Arun helped **Ravi** find **the book**. Later, **Ravi and Arun** discussed **the book** before leaving.”

f) Ambiguity if “He was looking for a book” Appeared Before Arun Was Introduced

If “He was looking for a book” appeared before Arun was introduced, the pronoun **“He” could only refer to an entity already introduced, such as Ravi**. If no male antecedent had been introduced, “He” would have an unresolved reference, making the sentence ambiguous or grammatically incomplete from a discourse perspective.

Q2. Dialog Act Recognition and Response Generation

The question requires a **Reassure + Advise** dialog act, entity coherence, a Cause–Effect or Elaboration relation, at least two required keywords, and a positive and polite tone.

a) Required Dialog Acts

- **Reassure**
- **Advise**

b) Important Entities

- Student
- Assignment
- Mistakes
- You
- Review
- Improvement/feedback

c) Three Possible Responses

Response 1:
“Don’t worry; you can **review** your assignment and identify any mistakes you may have made. This will help you **improve** your work and feel more **confident**.”

Response 2:
“It is normal to worry after submitting an assignment, so try to stay **confident**. You can use the **feedback** you receive to **improve** your future assignments.”

Response 3:
“You have already submitted your assignment, so there is no need to worry too much. When you receive **feedback**, **review** your mistakes carefully and use them to improve next time.”

d) Comparison

Response	Coherence	Constraint Satisfaction	Politeness	Relevance
Response 1	High	High	High	High
Response 2	High	High	High	High
Response 3	High	High	High	High

e) Best Response

Response 1 is the best response because it directly reassures the student, gives practical advice, maintains the entities “assignment” and “mistakes,” uses the required keywords **review, improve, and confident**, and clearly expresses a cause–effect relationship: reviewing mistakes helps improve the work and increases confidence.

f) If Entity Coherence Is Not Maintained

If entity coherence is not maintained, the response may become confusing because the system could lose track of who “you” refers to or what “it” and “mistakes” represent. This can produce an irrelevant or contradictory response and reduce the overall quality of the conversation.

Q3. Word Sense Disambiguation and Meaning Representation

The source sentence is: **"Priya sat on the bank and watched the boats moving across the water."**

a) Possible Meanings of "Bank"

The word **bank** has two possible meanings:

1. **Bank – Financial institution**
2. **Bank – Side of a river**

b) Correct Sense and Justification

The correct sense is **river bank**.

The words **"boats"** and **"water"** provide strong contextual evidence. Boats move across water, and Priya can sit on the side of a river and watch them. Therefore, the financial-institution meaning is not semantically compatible with the context.

c) Predicate Logic Representation

Let:

- priya = Priya
- b = river bank
- z = boat
- w = water

A suitable representation is:

sit(priya, b)
river_bank(b)
beside(priya, b)
watch(priya, z)
boat(z)
move(z)
water(w)
across(z, w)

Combined:

∃b ∃z ∃w [sit(priya,b) ∧ river_bank(b) ∧ beside(priya,b) ∧ boat(z) ∧ watch(priya,z) ∧ move(z) ∧ water(w) ∧ across(z,w)]

This represents that Priya is sitting beside the river bank and watching boats moving across the water.

d) Simple English Paraphrase

“Priya sat beside the riverbank and watched the boats moving across the water.”

This paraphrase removes the ambiguity of the word “bank.”

e) How WSD Improves Machine Translation

Word Sense Disambiguation improves Machine Translation by identifying the correct meaning of an ambiguous word from its context before translation. In this sentence, WSD identifies “bank” as a **riverbank**, not a financial institution. This allows the translation system to select the correct target-language word and produce a more accurate and coherent translation.

Rubrics for Calculation

Criteria	Weightage	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Problem Understanding	25%	Clearly defines the problem and identifies all relevant constraints accurately	Identifies most constraints with minor gaps	Partial understanding; misses key constraints	Fails to identify constraints correctly
Constraint Analysis	25%	Analyzes constraints effectively and prioritizes them logically	Provides reasonable analysis with some prioritization	Limited analysis; weak prioritization	No meaningful analysis or prioritization
Solution Development	25%	Develops optimal and feasible solutions satisfying all constraints	Develops workable solutions with minor limitations	Solutions partially satisfy constraints	Solutions do not meet constraints
Application of Concepts	15%	Effectively applies appropriate concepts, methods, or tools	Applies concepts with minor errors	Limited application; requires guidance	Unable to apply relevant concepts
Justification	10%	Provides strong justification for decisions with logical reasoning	Provides justification with some clarity	Weak or unclear justification	No justification provided

Q. No. / Criteria Level	Problem Understanding	Constraint Analysis	Solution Development	Application of Concepts	Justification	Sub. Total	Total %
1							
2							
3							

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Date: ____________________	Date: ____________________	Date: ____________________